

Cucumber: Cucumber (<https://cucumber.io/>) allows automated acceptance testing by executing the examples that optimally describe the behaviour of the application. It follows the Gherkin syntax form for creating scenarios. With more than 30 million downloads, it is the world's No. 1 tool for BDD adoption. It has cross-platform OS support and compatibility with most of the programming languages like Ruby, Java, Python and .NET.

Gauge: Gauge (<https://gauge.org/>) is an open source test automation framework released by ThoughtWorks, the creator of Selenium. It allows users to create readable and maintainable tests using the languages of your choice. It uses the Markdown syntax format for creating test specifications. Gauge has cross-platform and language support for writing test code. It supports Web, mobile and API interfaces for testing.

API automation testing tools

API testing is becoming critical since release cycles are becoming shorter with the adoption of agile and DevOps practices. With the 'shift left' testing approach, API testing bridges the gap between unit (application logic) and GUI layer testing.

Postman: Postman (<https://www.getpostman.com/>) is a good choice for API testing for those who don't want to deal with coding in an integrated development environment using the same language as the developers. Its characteristics are:

- Easy-to-use REST client
- Rich interface which makes it easy to use
- Can be used for both automated and exploratory testing
- Compactable on Mac, Windows, Linux and Chrome apps
- Has a bunch of integrations like support for Swagger and RAML formats
- Has run, test, document and monitoring features

Rest-Assured: Rest-Assured (<http://rest-assured.io/>) is an open source

Java domain-specific language that makes testing REST services simpler. It has many inbuilt functionalities, which means users do not have to code things from scratch. It integrates seamlessly with the Serenity automation framework, so that users can combine the UI and REST tests in one framework to generate awesome reports.

Mobile test automation tools

Appium: Appium (<http://appium.io/>) can help to automate workflows for native, hybrid and Web apps. It works on the client-server architecture. Appium can automate workflows of iOS, Android and Windows apps. It provides support for multiple programming languages like Java, Ruby, Python, PHP, JavaScript and C#. The popularity of this mobile automation testing tool is attributable to its easy installation and usage.

Robotium: Robotium (<http://robotium.com/>) is an open source test automation framework used for Android UI testing. It supports gray box UI testing, system testing, functional testing and user acceptance testing for both native and hybrid Android based applications.

Katalon: Katalon (<https://www.katalon.com/>) is a free Web, API and mobile test automation solution. It is simple to use, provides cross-platform support, and has good JIRA integration. You can create tests, run tests, view reports and maintain tests very easily using Katalon studio. Katalon adoption has been rising in recent years, and it was picked as the customers' choice in a survey done by Gartner in 2019.

Performance testing tools

Apache JMeter: Apache JMeter (<http://jmeter.apache.org/>) is an open source Java desktop app, intended for load testing of Web applications. It has many good features like dynamic reporting, portability, powerful test IDEs, etc. It supports different types of

applications, protocols, shell scripts, Java objects and databases.

K6: K6 (<https://k6.io/>) is an open source load and performance testing tool for testing cloud native applications, APIs and microservices. It's a modern developer-centric CLI tool with test cases written in ES6 JavaScript, and with built-in support for HTTP/1.1, HTTP/2 and Web Socket protocols. K6 can easily be introduced into automation pipelines in Jenkins, GitLab, Azure DevOps, CircleCI and other CI/CD tools for performance regression testing.

Grinder: Grinder (<http://grinder.sourceforge.net/>) is an open source Java based load testing framework. It uses multiple load injector machines, which make running a distributed test quite easy. Its main features include a generic approach, flexible scripting, distributed framework and mature http support.

Reliability and stability testing tools

Netflix Simian Army: Simian Army (<https://github.com/Netflix/SimianArmy/wiki>) consists of services (monkeys) in the cloud for generating various kinds of failures, detecting abnormal conditions and testing the ability to survive them. The goal is to keep the cloud safe, secure and highly available. Currently, the simians include Chaos Monkey, Janitor Monkey and Conformity Monkey.

Litmus: Litmus (<https://github.com/litmuschaos/litmus>) is a toolset to do chaos engineering in a Kubernetes native way. Litmus provides chaos CRDs (custom resource definitions) for cloud native developers and site reliability engineers to inject, orchestrate and monitor chaos to find weaknesses in Kubernetes deployments.

Accessibility testing tools

As per World Bank estimates, around 15 per cent of the world population experiences some form of disability. Several countries have implemented legislation that makes it mandatory for organisations to ensure their products